



## CHEMISTRY (Matter)

**MATTER** – anything that occupies space and has mass

### Properties of Matter:

- 1. Chemical Properties** – properties that can only be observed when the substance produced a new material (burning, rusting)
- 2. Physical Properties** – properties that can be perceived by the physical senses
  - a) Extensive:** depends on mass of substance (weight, volume, height)
  - b) Intensive:** does not depend on mass but on type of substance (density, color, odor, boiling point)

### PHASES OF MATTER:

| Property                       | Solid                                    | Liquid                            | Gas                       |
|--------------------------------|------------------------------------------|-----------------------------------|---------------------------|
| <b>Shape</b>                   | Keeps own shape                          | Takes shape of container          | Takes shape of container  |
| <b>Volume</b>                  | Keeps own volume                         | Keeps own volume                  | Takes volume of container |
| <b>Density</b>                 | High to very high                        | Medium to high                    | Low                       |
| <b>Can be compressed</b>       | Not at all                               | Only a little                     | A great deal              |
| <b>Force between particles</b> | Strong                                   | Strong                            | Weak                      |
| <b>Motion of particles</b>     | Vibrate fixed with respect to each other | Vibrate move within liquid volume | Vibrate move anywhere     |

### CLASSIFICATION OF MATTER:

#### 1. PURE SUBSTANCES

Pure type matter which cannot be separated physically. Either a compound or an element.

| ELEMENT                                                      | COMPOUND                                                                                                                |
|--------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Substances that cannot be separated into simpler components. | Combination of elements that can be separated by chemical means. Elements combine to form compounds to attain stability |

|                   |                                                   |             |                                                                             |
|-------------------|---------------------------------------------------|-------------|-----------------------------------------------------------------------------|
| <b>Metal</b>      | Conductors, malleable, ductile, lustrous          | <b>Acid</b> | pH <7; turns blue litmus paper to red gives hydronium ion $H_3O^+$ in water |
| <b>Non-metal</b>  | Insulators, non malleable, brittle, non lustrous  | <b>Base</b> | pH >7; turns red litmus paper to blue; gives hydroxyl ion $OH^-$ in water   |
| <b>Metalloids</b> | Exhibit the properties of a metal and a non metal | <b>Salt</b> | Product of a reaction of an acid and a base                                 |

#### 2. MIXTURES

Combinations of two or more substances

| Heterogenous                                                                                                                                                |                                                                                                      | Homogenous                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Observed with different phases                                                                                                                              |                                                                                                      | Observed to exhibit in one phase; can be physically separated |
| <b>Colloid</b>                                                                                                                                              | <b>Suspension</b>                                                                                    |                                                               |
| Solutes are dispersed throughout the solution and particles do not settle. Particles are bigger than that of solution but smaller than that of suspensions. | Mixture in which the particles are big enough to be seen by the naked eye and settles in the bottom. |                                                               |

#### METHODS OF SEPARATING MIXTURES

- 1. Filtration** – separating suspended solid from liquid by passing liquid through a filter
- 2. Decantation** – separating two liquids with different densities by pouring out less dense liquid
- 3. Evaporation** – separating solid from liquid by turning the liquid into vapor
- 4. Distillation** – separating two liquids with different boiling points by means of heat



5. **Centrifugation** – sedimentation of mixtures by use of the centrifugal force
6. **Magnets** – separate metals from non metals by the use of magnets

### CHANGES THAT MATTER UNDERGOES

1. **Physical** – changes in the physical properties of matter and takes place without changing the chemical composition
2. **Chemical** – results after a chemical reaction where a new substance is formed
3. **Nuclear** – changes in the atomic properties of an element
4. **Phase Change** – change in the state of matter (solid, liquid or gas)
  - **Exothermic process:** process where heat is expelled such as FREEZING and CONDENSATION
  - **Endothermic process:** processes where heat is absorbed such as MELTING and VAPORIZATION
  - **Enthalpy:** rate of disorder the higher the enthalpy, the more disordered the particles

**COMPOSITION OF MATTER:** matter is composed of tiny particles called atoms.

**ATOMS** – made up of:

- Protons (+ charge)
- Neutrons (neutral charge)
- Electrons (- charge)

**NUCLEUS** – dense center of the atom made of protons and nucleus.

- Mass of proton is almost equal to mass of neutron
- Electrons surround the nucleus

**MASS NUMBER/ ATOMIC MASS** – sum of the number of protons and the number of neutrons

- concentrated in the nucleus

**ATOMIC NUMBER** – number of protons in the element

- Differentiates atoms of different elements
- Does not change even if particle has charge of none

**Example:** All atoms containing 1 proton, have 1 atomic number and are all hydrogen.

All atoms containing 6 protons have 6 atomic number and are all carbon.

### REMEMBER:

|                        |                                                          |
|------------------------|----------------------------------------------------------|
| <b>Atomic mass (A)</b> | A = protons + neutrons<br>A = Atomic no. (Z) + neutrons  |
| <b>No. of Neutrons</b> | No. of Neutrons = A - Z                                  |
| <b>No. of Protons</b>  | No. of Protons = electrons (for neutrally charged atoms) |

### Chemical Symbol

- letter representation of an element is  ${}^A_ZX$  where X is the hypothetical element.

**Example:** The symbol  ${}^{35}_{17}\text{Cl}$  refers to chlorine atoms with the following particles:

**Atomic No. (Z)** = protons = electrons = 17

**Atomic mass (A)** = 35

**No. of Neutrons** =  $35 - 17 = 8$

**ISOTOPE:** elements have the same number of protons but different number of neutrons

**Example:**  ${}^{12}_6\text{C}$ ,  ${}^{11}_6\text{C}$ ,  ${}^{13}_6\text{C}$

### CHEMICAL NOMENCLATURE

- **ION** – atoms with unequal number of protons and electrons, as electrons are lost or added.
- **CATIONS** – positively charged atoms (number of electrons is less than the number of protons)  
**Example:**  $\text{Na}^+$ ,  $\text{Ca}^{2+}$
- **ANIONS** – negatively charged atoms (number of electrons is greater than the number of protons)  
**Example:**  $\text{Cl}^-$ ,  $\text{F}^-$ ,  $\text{Br}^-$

### REMEMBER:

**Charge of Ion** = protons – electrons = Z – electrons

**Example:** Determine number of particles of  $\text{Br}^-$  with atomic mass of 80

**Solution:** Bromine has an atomic number of 35 which is constant despite changes in charge; the number of protons is equal to the atomic number so Number of protons = 35. From the symbol, it is known that it has -1 charge.

**Charge of ion** = protons – electrons = Z – electrons

$$-1 = 35 - \text{electrons}$$

$$-1 - 35 = \text{electrons}$$

$$36 = \text{electrons}$$

**Atomic mass (A)** = atomic no. (Z) + neutrons

$$80 = 35 + \text{neutrons}$$

$$80 - 35 = \text{neutrons}$$

$$45 = \text{neutrons}$$